

Project Proposal

DiffSplat: Reproducing a Scalable 3D Gaussian Splat Generation Framework

Generative Deep Learning | Group 6

Kaustubh Shah, Sree Gnanesh Dommeti, Nishchal Marur Nanjunda Swamy, Haochen Yang

1. Introduction

The generation of three-dimensional content from text or image inputs has rapidly matured into one of the most active areas in generative deep learning. Early approaches based on score distillation sampling, such as DreamFusion [5], relied on optimizing a 3D representation per scene using a frozen 2D diffusion prior, resulting in generation times on the order of hours per object. More recent feed-forward methods have shifted toward amortized generation, training a model that directly predicts 3D representations in a single forward pass by leveraging the rich priors encoded in large-scale image diffusion models.

DiffSplat [4], accepted at ICLR 2025, represents a leading instantiation of this paradigm. It fine-tunes pretrained text-to-image diffusion models to natively generate structured 3D Gaussian splat grids, achieving 3D generation in one to two seconds. The model introduces a lightweight reconstruction module for scalable data curation and a joint diffusion-plus-rendering loss to enforce 3D consistency across arbitrary viewpoints.

This project proposes to reproduce the DiffSplat pipeline in full, validating its core design choices and benchmarking its performance on text-conditioned and image-conditioned 3D generation tasks. Through careful reproduction, we aim to develop a deep understanding of how large-scale image diffusion priors can be repurposed for native 3D generation, and to identify the key components that drive the model's performance.

2. Problem Statement

The central problem addressed in DiffSplat, and in this project, is: how can a pretrained 2D image diffusion model be repurposed to generate geometrically consistent 3D content without training a 3D generative model from scratch? Prior approaches either train native 3D networks on scarce 3D datasets, suffer from multi-view inconsistency due to independent 2D generation, or require per-scene optimization that is computationally prohibitive.

DiffSplat addresses this by adopting 3D Gaussian Splatting [3] as the scene representation and fine-tuning a pretrained image diffusion model to denoise structured Gaussian splat grids rather than pixel grids. The training objective combines a standard diffusion loss on splat latents with a 3D rendering loss that enforces multi-view coherence across arbitrary camera poses. This project will reproduce this pipeline end-to-end and validate that the combination of these two loss terms is indeed necessary for high-quality generation, as claimed in the original ablation studies.

3. Related Work

This project builds on four core lines of prior work:

- **[1] Denoising Diffusion Probabilistic Models (Ho et al., 2020):** Established the theoretical and practical foundation for diffusion-based generative modeling. DiffSplat's denoising training objective is a direct extension of the DDPM framework to the domain of Gaussian splat latents.
- **[2] Latent Diffusion Models (Rombach et al., 2022):** Introduced the latent diffusion framework in which a diffusion model operates in the compressed latent space of a pretrained autoencoder. DiffSplat extends this by fine-tuning the image VAE to also encode Gaussian splat properties, enabling a shared latent space between image and splat representations.
- **[3] 3D Gaussian Splatting (Kerbl et al., 2023):** Proposed an explicit differentiable 3D scene representation based on anisotropic Gaussians, rendered via tile-based rasterization. DiffSplat adopts this representation for its efficient rendering and quality balance, and uses its differentiable renderer to compute the 3D rendering loss during training.
- **[4] DiffSplat (Lin et al., 2025):** The primary anchor paper for this project. DiffSplat introduces structured Gaussian splat grids as a 3D representation suitable for direct generation by image diffusion models, proposes a lightweight reconstruction model for scalable dataset curation, and demonstrates state-of-the-art performance on text-conditioned and image-conditioned 3D generation tasks.

4. Approach

4.1 Reproducing the DiffSplat Pipeline

The reproduction will follow the three-stage architecture described in the paper. In the first stage, we will set up and run the structured Gaussian reconstruction module (GSRecon), a lightweight feed-forward network that takes multi-view posed images as input and regresses per-pixel Gaussian splat properties in under 0.1 seconds. This module is used at training time to generate pseudo ground-truth splat grids from the G-Objaverse dataset for use as training targets.

In the second stage, we will fine-tune and evaluate the Gaussian VAE (GSVAE), which extends the image VAE from the base diffusion model to also encode and decode Gaussian splat properties. The key design choice here is that the splat latents are kept in the same latent space as image latents, enabling the pretrained image diffusion model to be applied with minimal architectural modification.

In the third stage, we will run inference with the full DiffSplat generative model under both text and image conditioning. The model generates structured splat grids by denoising splat latents using the fine-tuned diffusion backbone, then decodes them through the GSVAE and renders novel views using the 3DGS rasterizer.

4.2 Ablation Studies

Beyond basic reproduction of inference results, we will reproduce the key ablation studies reported in the paper to validate our understanding of the model's design. Specifically, we will evaluate the effect of removing the 3D rendering loss (setting the rendering loss weight to zero) to confirm that the diffusion loss alone is insufficient for 3D-consistent generation, and the effect of removing the diffusion loss to confirm that treating splat latents as a native 3D representation

degrades quality. These ablations are central to the paper's claims and form the backbone of our reproduction effort.

4.3 Evaluation

Quantitative evaluation will follow the protocol established in the paper. For text-conditioned generation, we report CLIP similarity score and CLIP R-Precision using the ViT-B/32 backbone, and ImageReward to measure human aesthetic preference, evaluated on the T3Bench prompt set. For image-conditioned generation and reconstruction, we report PSNR, SSIM, and LPIPS on held-out objects from the Google Scanned Objects (GSO) dataset. Results will be compared against the numbers reported in the DiffSplat paper to assess the fidelity of our reproduction.

5. Scope and Deliverables

The scope of this project is defined as follows:

- **In scope:** Full reproduction of DiffSplat inference under text and image conditioning using the official codebase and pretrained checkpoints; reproduction of the key ablation studies on the diffusion and rendering loss weights; quantitative evaluation on the T3Bench and GSO benchmarks; qualitative comparison of rendered novel views; final written report and oral presentation.
- **Out of scope:** Full retraining of DiffSplat or its component models from scratch; training the GSRecon or GSVAE modules; video or dynamic scene generation; novel architectural modifications; real-time deployment.

6. References

- [1] Ho, J., Jain, A., & Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. *Advances in Neural Information Processing Systems (NeurIPS)*, 33. <https://arxiv.org/abs/2006.11239>
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- [3] Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4). <https://arxiv.org/abs/2308.04079>
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- [5] Poole, B., Jain, A., Barron, J. T., & Mildenhall, B. (2022). DreamFusion: Text-to-3D using 2D Diffusion. *International Conference on Learning Representations (ICLR 2023)*. <https://arxiv.org/abs/2209.14988>